

Unsupervised Phenotyping for Diabetes Mellitus Type 2

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Research Project Introduction

In 2018, Swedish scientists identified subgroups for adult-onset diabetes (Type 2) that can help to refine treatment plans and identify risk-complex individuals (Ahlqvist et al., 2018). The researchers utilized data from the Swedish All New Diabetics in Scania cohort and performed k-means and hierarchical clustering to determine the subgroups. This project will expand the research to the United States and will use advanced unsupervised clustering techniques and explainable AI to determine if there are any novel or United States-specific phenotypes within Type 2 diabetes and their defining characteristics.

Hypotheses & Method

Hypothesis: There exists mathematically unique subgroups within diabetes Type 2 that did not exist in the previous research and may be unique to the United States.

Data and Methodology: The research will utilize the NHANES database to gather data for individuals diagnosed with diabetes Type 2 and will include biological and genomic markers. The individuals' records will be vectorized and clustered using Gaussian Mixture Models and UMAP for manifold projection, improving upon the prior research. A tree-based model will then be used to extract the defining features of each cluster.

Prior Research

Ahlqvist, E., Storm, P., Käräjämäki, A., Martinell, M., Dorkhan, M., Carlsson, A., Vikman, P., Prasad, R. B., Aly, D. M., Almgren, P., Wessman, Y., Shaat, N., Spégel, P., Mulder, H., Lindholm, E., Melander, O., Hansson, O., Malmqvist, U., Lernmark, Å., Lahti, K., ... Groop, L. (2018). Novel subgroups of adult-onset diabetes and their association with outcomes: a data-driven cluster analysis of six variables. *The lancet. Diabetes & endocrinology*, 6(5), 361–369. [https://doi.org/10.1016/S2213-8587\(18\)30051-2](https://doi.org/10.1016/S2213-8587(18)30051-2)

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Manzini, E., Vlachos, B., Franch-Nadal, J., Escudero, J., Génova, A., Reixach, E., Andrés, E., Pizarro, I., Portero, J.-L., Mauricio, D., & Perera-Lluna, A. (2022). Longitudinal deep learning clustering of Type 2 Diabetes Mellitus trajectories using routinely collected health records. *Journal of Biomedical Informatics*, 135, 104218. <https://doi.org/10.1016/j.jbi.2022.104218>